

Course

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3.1.1. Numpy Array Operations

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Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using `ndim`
- The shape of the array using `shape`
- The total number of elements in the array using `size`

Assume all Input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, `r` and `c`, representing the number of rows and the number of columns.
- The next `r` lines contain `c` space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Note: Use `reshape()` function to reshape the Input array with the specified number of rows and columns.

Sample Test Cases

numpyarr...

Submit

1

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import numpy as np

r,c = map(int,input().split())

data=[]

if r==0 and c==0:

arr=np.empty((0,0),dtype=int)

else:

for _ in range(r):

row=list(map(int,input().split()))

data.append(row)

arr=np.array(data)

print(arr)

print(arr.ndim)

print(arr.shape)

print(arr.size)

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3.2.1. Numpy: Matrix Operations

The given code takes two 3×3 matrices, `matrix_a`, and `matrix_b`, as input from the user and converts them into NumPy arrays.

Task:
You are required to compute and display the results of the following matrix operations:
1. **Addition** (`matrix_a + matrix_b`)
2. **Subtraction** (`matrix_a - matrix_b`)
3. **Element-wise Multiplication** (`matrix_a * matrix_b`)
4. **Matrix Multiplication** (`matrix_a · matrix_b`)
5. **Transpose of Matrix A**

Input Format:
• The user will input 3 rows for `matrix_a`, each containing 3 integers separated by spaces.
• Similarly, the user will input 3 rows for `matrix_b`, each containing 3 integers separated by spaces.

Output Format:
The program should display the results of the operations in the following order:
1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

Sample Test Cases

matrixOp...

1 `import numpy as np`
2
3 `# Input matrices`
4 `print("Enter Matrix A:")`
5 `matrix_a = np.array([list(map(int, input().split())) for i in range(3)])`
6
7 `print("Enter Matrix B:")`
8 `matrix_b = np.array([list(map(int, input().split())) for i in range(3)])`
9
10
11 `# Addition`
12 `print("Addition (A + B):")`
13 `print(matrix_a + matrix_b)`
14 `# Subtraction`
15 `print("Subtraction (A - B):")`
16 `print(matrix_a - matrix_b)`
17 `# Multiplication (element-wise)`
18 `print("Element-wise Multiplication (A * B):")`
19 `print(matrix_a * matrix_b)`
20 `# Matrix multiplication (dot product)`
21 `print("A dot B:")`
22 `print(matrix_a .dot( matrix_b))`
23 `# Transpose`
24 `print("Transpose of A:")`
25 `print(matrix_a.T)`

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3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays

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You are given two arrays `arr1` and `arr2`. You need to perform horizontal and vertical stacking operations on them using NumPy.

- Horizontal Stacking: Stack the two matrices horizontally (side by side).
- Vertical Stacking: Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Sample Test Cases

stacking.py

Submit

1 import numpy as np

2

3 # Input matrices

4 print("Enter Array1:")

5 arr1 = np.array([list(map(int, input().split())) for i in range(3)])

6

7 print("Enter Array2:")

8 arr2 = np.array([list(map(int, input().split())) for i in range(3)])

9

10 # Perform horizontal stacking (hstack)

11 result=np.hstack((arr1,arr2))

12 print("Horizontal Stack:")

13 print(result)

14

15

16 # Perform vertical stacking (vstack)

17 result=np.vstack((arr1,arr2))

18 print("Vertical Stack:")

19 print(result)

20

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3.2.3. Numpy: Custom Sequence Generation

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using `numpy` based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

Sample Test Cases

customS...

1 `import numpy as np`

2

3 `# Take user input for the start, stop, and step of the sequence`

4 `start = int(input())`

5 `stop = int(input())`

6 `step = int(input())`

7

8 `# Generate the sequence using np.arange()`

9 `sequence = np.arange(start, stop, step)`

10 `# Print the generated sequence`

11 `print(sequence)`

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3.2.4. Numpy: Arithmetic and Statistical Operations, Mathematical Op...

You are given two arrays A and B. Your task is to complete the function array_operations, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:

• Compute the element-wise sum, difference, and product of the two arrays.

2. Statistical Operations:

• Calculate the mean, median, and standard deviation of array A.

3. Bitwise Operations:

• Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: A_i OR B_i).

Input Format:

• The first line contains space-separated integers representing the elements of array A.

• The second line contains space-separated integers representing the elements of array B.

Output Format:

• For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

Sample Test Cases

different...

1 import numpy as np

2

3 def array_operations(A, B):

4

5 # Convert A and B to NumPy arrays

6 arr_A=np.array(A)

7 arr_B=np.array(B)

8 # Arithmetic Operations

9 sum_result = arr_A + arr_B

10 diff_result = arr_A - arr_B

11 prod_result = arr_A * arr_B

12

13 # Statistical Operations

14 mean_A = np.mean(arr_A)

15 median_A = np.median(arr_A)

16 std_dev_A = np.std(arr_A)

17

18 # Bitwise Operations

19 and_result = arr_A & arr_B

20 or_result = arr_A | arr_B

21 xor_result = arr_A ^ arr_B

22

23 # Output results with one space between each element

24 print("Element-wise Sum:", ' '.join(map(str, sum_result)))

25 print("Element-wise Difference:", ' '.join(map(str, diff_result)))

26 print("Element-wise Product:", ' '.join(map(str, prod_result)))

27

28 print(f"Mean of A: {mean_A}")

29 print(f"Median of A: {median_A}")

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3.2.5. Numpy: Copying and Viewing Arrays

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The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

Creating a view of the original_array and assigning it to view_array.

Creating a copy of the original_array and assigning it to copy_array.

After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.

Input Format:

A single line of space-separated integers.

Output Format:

After modifying the view:

Original array after modifying view: <original_array>
View array: <view_array>

After modifying the copy:

Original array after modifying copy: <original_array>
Copy array: <copy_array>

Sample Test Cases

copyAnd...

Submit

1 import numpy as np

2

3 inputlist = list(map(int,input().split(" ")))

4

5 # Original array

6 original_array = np.array(inputlist)

7

8 # Create a view

9 view_array =original_array.view()

10

11 # Create a copy

12 copy_array =original_array.copy()

13

14 # Modify the view

15 view_array[0] = 99

16 print("Original array after modifying view:", original_array)

17 print("View array:", view_array)

18

19 # Modify the copy

20 copy_array[1] = 88

21 print("Original array after modifying copy:", original_array)

22 print("Copy array:", copy_array)

23

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3.2.6. Numpy: Searching, Sorting, Counting, Broadcasting

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The given code in the editor takes a single array, array1, as space-separated integers as input from the user.
Additionally, it takes the following inputs:

- search_value: The value to search for in the array.
- count_value: The value to count its occurrences in the array.
- broadcast_value: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

- Searching:** Find the indices where search_value appears in array1 and print these indices.
- Counting:** Count how many times count_value appears in array1 and print the count.
- Broadcasting:** Add broadcast_value to each element of array1 using broadcasting, and print the resulting array.
- Sorting:** Sort array1 in ascending order and print the sorted array.

Input Format:

- A single line containing space-separated integers representing array1.
- An integer search_value represents the value to search for in the array.
- An integer count_value represents the value to count in the array.
- An integer broadcast_value represents the value to add to each element of the array.

Output Format:

- The indices where search_value occurs in array1.
- The count of occurrences of count_value in array1.
- The array after adding the broadcast_value to each element.
- The sorted array.

Sample Test Cases

arrayOpe...

Submit

```
1 import numpy as np
2
3 # Input array from the user
4 array1 = np.array(list(map(int, input().split())))
5
6 # Searching
7 search_value = int(input("Value to search: "))
8 count_value = int(input("Value to count: "))
9 broadcast_value = int(input("Value to add: "))
10
11 # Find indices where value matches in array1
12 indices = np.where(array1 == search_value)[0]
13 print(indices)
14 # Count occurrences in array1
15 count = np.count_nonzero(array1 == count_value)
16 print(count)
17 # Broadcasting addition
18 broadcasted_array = array1 + broadcast_value
19 print(broadcasted_array)
20 # Sort the first array
21 sorted_array = np.sort(array1)
22 print(sorted_array)
```

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Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details:** Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students:** Determine the total number of students in the dataset.
- **Print all student roll numbers:** Extract and print the roll numbers of all students.
- **Print Subject 1 marks:** Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2:** Identify the lowest marks in Subject 2.
- **Find maximum marks in Subject 3:** Identify the highest marks in Subject 3.
- **Print all subject marks:** Display the marks of all students for each subject.
- **Find total marks of students:** Compute the total marks for each student across all subjects.
- **Find the average marks of each student:** Compute the average marks for each student.
- **Find average marks of each subject:** Compute the average marks for all students in each subject.
- **Find average marks of Subject 1 and Subject 2:** Compute the average marks for Subject 1 and Subject 2.
- **Find average marks of Subject 1 and Subject 3:** Compute the average marks for Subject 1 and Subject 3.
- **Find the roll number of the student with maximum marks in Subject 3:** Identify the student with the highest marks in Subject 3 and print their roll number.
- **Find the roll number of the student with minimum marks in Subject 2:** Identify the student with the lowest marks in Subject 2 and print their roll number.
- **Find the roll number of students who scored 24 marks in Subject 2:** Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- **Find the count of students who got less than 40 marks in Subject 1:** Count the number of students who scored less than 40 marks in Subject 1.
- **Find the count of students who got more than 90 marks in Subject 2:** Count the number of students who scored more than 90 marks in Subject 2.

Operation...

```
1 import numpy as np
2
3 a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)
4
5 # All student details
6 print("All student Details:")
7 print(" "+str(a)) -- # FIX 1: correct spacing (no underscore issue)
8
9 # Total students
10 print("Total Students:", a.shape[0])
11
12 # Roll numbers
13 roll = a[:,0]
14 print("All Student Roll Nos", roll)
15
16 # Subject 1
17 s1 = a[:,1]
18 print("Subject 1 Marks", s1)
19
20 # Subject 2 & 3
21 s2 = a[:,2]
22 s3 = a[:,3]
23
24 print("Min marks in Subject 2", np.min(s2))
25 print("Max marks in Subject 3", np.max(s3))
26
27 # Subjects
28 subjects = a[:,1:]
29 print("All subject marks:", subjects)
30
```

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